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Final Exam

AP Calc AB

FINAL EXAM · AP CALCULUS AB

The Whole

35 questions covering every section of the study guide — limits, derivatives, integrals, applications, and physics. Q1-29 multiple choice, Q30 multi-part numerical, Q31-35 free response. Click an option to select it (you can change your mind). When you're finished, hit Copy My Answers at the bottom to grab everything formatted for Canvas or email.

29 Multiple Choice 1 Numerical (4 parts) 5 Free Response

Submit by Copy

SECTION A · MULTIPLE CHOICE (1-29)

Question 1

FOUNDATIONS · LIMITS

Evaluate: $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

A 0

B 3

C ∞

D 6

E 9

F Does not exist

Question 2

FOUNDATIONS · CONTINUITY

For $f(x) = (x^2 - 4)/(x - 2)$, what type of discontinuity exists at $x = 2$?

A Removable (hole)

B Jump

C Infinite (asymptote)

D Oscillating

E f is continuous at $x = 2$

F Endpoint discontinuity

Question 3

FOUNDATIONS · LIMITS AT ∞

Evaluate: $\lim_{x\to\infty} \frac{3x^2 + 5x - 1}{6x^2 - 2x + 4}$

A 0

B 1/3

C 2

D 3

E 1/2

F ∞

Question 4

FOUNDATIONS · UNIT CIRCLE

Find: $\sin(7\pi/6)$

A $1/2$

B $\sqrt{3}/2$

C $-\sqrt{3}/2$

D $\sqrt{2}/2$

E $-\sqrt{2}/2$

F $-1/2$

Question 5

THEOREMS · IVT

A function f is continuous on $[1, 5]$ with $f(1) = -2$ and $f(5) = 8$. By the Intermediate Value Theorem, which value of k is GUARANTEED to have a corresponding c in $(1, 5)$ where $f(c) = k$?

A $k = -5$

B $k = 4$

C $k = 9$

D $k = 12$

E $k = -3$

F None of the above

Question 6

THEOREMS · MVT

A function f is differentiable on $[2, 8]$ with $f(2) = 4$ and $f(8) = 22$. By the Mean Value Theorem, there exists c in $(2, 8)$ such that $f'(c) =$

A 1

B 2

C 6

D 9

E 3

F 18

Question 7

DIFFERENTIATION · POWER RULE

If $f(x) = 4x^3 - 2x^2 + 7x - 5$, find $f'(x)$.

A $4x^2 - 2x + 7$

B $12x^2 - 2x + 7$

C $12x^3 - 4x^2 + 7x$

D $x^3 - x^2 + 7$

E $4x^2 - 4x + 7$

F $12x^2 - 4x + 7$

Question 8

DIFFERENTIATION · PRODUCT RULE

If $f(x) = x^2 \cdot \sin(x)$, find $f'(x)$.

A $2x \cos(x)$

B $2x \cos(x) - x^2 \sin(x)$

C $2x \sin(x) - x^2 \cos(x)$

D $2x \sin(x) + x^2 \cos(x)$

E $x^2 \cos(x)$

F $2x^2 \cos(x)$

Question 9

DIFFERENTIATION · QUOTIENT RULE

If $f(x) = (3x + 1)/(x - 2)$, find $f'(x)$.

A $-7/(x - 2)^2$

B $3/(x - 2)^2$

C $(3x + 1)/(x - 2)^2$

D $-5/(x - 2)^2$

E $7/(x - 2)^2$

F $(6x - 7)/(x - 2)^2$

Question 10

DIFFERENTIATION · CHAIN RULE

If $f(x) = \sin(3x^2)$, find $f'(x)$.

A $\cos(3x^2)$

B $\cos(6x)$

C $3x \cos(3x^2)$

D $-6x \sin(3x^2)$

E $6x \sin(3x^2)$

F $6x \cos(3x^2)$

Question 11

DIFFERENTIATION · TRIG

What is the derivative of $f(x) = \tan(x)$?

A $\sec(x)$

B $\sec(x) \tan(x)$

C $-\csc^2(x)$

D $\sec^2(x)$

E $\cot(x)$

F $1/\cos(x)$

Question 12

DIFFERENTIATION · LOG

If $f(x) = \ln(5x^2 + 1)$, find $f'(x)$.

A $1/(5x^2 + 1)$

B $10x/(5x^2 + 1)$

C $(5x^2 + 1)/(10x)$

D $10x$

E $1/(10x)$

F $5x^2/(5x^2 + 1)$

Question 13

IMPLICIT DIFFERENTIATION

For the curve $x^2 + y^2 = 25$, find dy/dx.

A x/y

B $-y/x$

C $-x/y$

D y/x

E $-2x/y$

F $2y/x$

Question 14

IMPLICIT · PRODUCT TERM

For the curve $xy + y^2 = 6$, find dy/dx.

A $y/(x + 2y)$

B $y/(x - 2y)$

C $-y/(2y)$

D $(1 + 2y)/y$

E $-y/(x + 2y)$

F $1/(x + 2y)$

Question 15

APP. OF DERIVATIVES · EXTREMA

For $f(x) = x^3 - 3x^2 - 9x + 5$, at which x-value does a **local minimum** occur?

A $x = -3$

B $x = 3$

C $x = -1$

D $x = 0$

E $x = 1$

F $x = 5$

Question 16

APP. OF DERIVATIVES · L'HÔPITAL

Evaluate: $\lim_{x \to 0} \frac{\sin(2x)}{x}$

A 0

B $1/2$

C 1

D ∞

E Does not exist

F 2

Question 17

APP. OF DERIVATIVES · RELATED RATES

A spherical balloon's radius is increasing at **2 cm/s**. When **r = 5 cm**, how fast is the volume changing? **[V = (4/3)πr³]**

A 25π cm³/s

B 50π cm³/s

C 100π cm³/s

D 200π cm³/s

E 400π cm³/s

F 500π/3 cm³/s

Question 18

APP. OF DERIVATIVES · CONCAVITY

For **f(x) = x⁴ − 4x³**, on which interval(s) is the graph **concave up**?

A (−∞, 0) ∪ (2, ∞)

B (0, 2)

C (−∞, 0) only

D (2, ∞) only

E All real numbers

F (0, ∞)

Question 19

ANTIDERIVATIVES · +C RULE

Find: $\int(6x^2 - 4x + 3)\,dx$

A $2x^3 - 2x^2 + 3x$ (no +C)

B $12x - 4 + C$

C $6x^3 - 4x^2 + 3x + C$

D $2x^3 + 2x^2 + 3x + C$

E $2x^3 - 2x^2 + 3x + C$

F $2x^2 - 2x + 3 + C$

Question 20

ANTIDERIVATIVES · TRIG

Find: $\int\sec^2(x)\,dx$

A $\sec(x) + C$

B $\sec(x)\tan(x) + C$

C $\tan(x) + C$

D $-\cot(x) + C$

E $-\csc^2(x) + C$

F $\ln|\sec(x)| + C$

Question 21

U-SUBSTITUTION

Find: $\int 2x \cdot \cos(x^2)\,dx$

A $\cos(x^2) + C$

B $2\sin(x^2) + C$

C $-\sin(x^2) + C$

D $\sin(x^2) + C$

E $x^2 \sin(x^2) + C$

F $\sin(x) \cdot \cos(x) + C$

Question 22

U-SUB · DEFINITE

Evaluate: $\int_0^2 3x^2 \cdot e^{x^3}\, dx$

A $e^8 - 1$

B $e^2 - 1$

C e^8

D $e^8 - e$

E $8 - e$

F $e^3 - 1$

Question 23

RIEMANN SUM · LEFT

Approximate $\int_0^4 f(x)\, dx$ using a **LEFT Riemann sum** with n = 4 subintervals ($\Delta x = 1$) using the table:

x	0	1	2	3	4
f(x)	5	7	8	10	12

A 26

B 28

C 32

D 35

E 30

F 37

Question 24

RIEMANN · OVER/UNDER

For an **increasing** and **concave up** function, a **LEFT** Riemann sum produces an estimate that is:

A An overestimate

B Always exact

C An overestimate if Δx is small

D Depends on the function

E Sometimes over, sometimes under

F An underestimate

Question 25

DEFINITE INTEGRALS · PROPERTIES

Given that $\int_0^3 f(x)\,dx = 7$ and $\int_3^5 f(x)\,dx = -2$, find $\int_0^5 f(x)\,dx$.

A −9

B 5

C −5

D 2

E 9

F 14

Question 26

DEFINITE INTEGRALS · GRAPHS

The graph of f consists of a **semicircle** of radius 3 above the x-axis from $x = -3$ to $x = 3$, plus a **triangle** below the x-axis with base 4 and height 2 from $x = 3$ to $x = 7$. Find $\int_{-3}^{7} f(x)\,dx$.

A $9\pi/2 + 4$

B $9\pi - 4$

C $9\pi/2$

D $4 - 9\pi/2$

E $9\pi + 4$

F $9\pi/2 - 4$

Question 27

FTC PART 1 · CHAIN RULE

If $g(x) = \int_{0}^{x^2} \sin(t)\,dt$, find $g'(x)$.

A $\sin(x^2)$

B $2x\cos(x^2)$

C $2x\sin(x^2)$

D $\cos(x^2) - 1$

E $\sin(x^2) - \sin(0)$

F $x^2\cos(x)$

Question 28

APP. OF INTEGRALS · AREA

Find the area of the region bounded by $y = 2x$ and $y = x^2$ from $x = 0$ to $x = 2$.

A $2/3$

B 1

C 8/3

D 4/3

E 2

F 4

Question 29

PHYSICS · SHM

A particle moves along the x-axis with position $x(t) = 4\cos(2t)$. What is its velocity at $t = \pi/4$?

A −4

B 0

C 4

D 8

E −8

F 8 sin(π/2)

SECTION B · MULTI-PART NUMERICAL (30)

Question 30

MULTI-PART · PARTICLE MOTION

A particle moves along the x-axis with velocity $v(t) = 3t^2 - 12t + 9$ for $0 \le t \le 5$.

Answer all four parts. Enter exact numerical values (no units).

PART (A)

The smaller value of t at which the particle is at rest:

Your answer.

PART (B)

The larger value of **t** at which the particle is at rest:

Your answer.

PART (C)

The total **displacement** from t = 0 to t = 5: $\int_0^5 v(t)\,dt$

Your answer.

PART (D)

The **acceleration** at t = 2: a(2)

Your answer.

SECTION C · FREE RESPONSE (31-35)

Question 31

FRQ · RELATED RATES

A 13-foot ladder leans against a vertical wall. The bottom of the ladder is sliding away from the wall at a rate of 2 ft/s.

(a) How fast is the top of the ladder sliding down the wall when the bottom is 5 ft from the wall?

(b) Justify why your answer to (a) must be negative, in context.

Show your work — set-up, work, and final answer with units.

Question 32

FRQ · MVT JUSTIFICATION

A function f is continuous on [2, 10] and differentiable on (2, 10), with the values:

T	2	4	7	10
f(t)	5	9	14	21

(a) Use the Mean Value Theorem to justify why there must exist a value **c** in **(2, 10)** such that **f'(c) = 2**.

(b) Use a left Riemann sum with the 3 subintervals shown to approximate $\int_2^{10} f(t)\,dt$.

Show your work — set-up, work, and final answer with units.

Question 33

FRQ · AREA & VOLUME

Let R be the region in the first quadrant bounded by **y = √x** and **y = x/2**.

(a) Find the x-coordinates of the points where the two curves intersect.

(b) Find the area of region R.

(c) Find the volume of the solid generated when R is rotated about the x-axis.

Show your work — set-up, work, and final answer with units.

Question 34

FRQ · GRAPH OF F'

The graph of $f'(x)$, the derivative of f, on $[0, 8]$ consists of two pieces:

- A **semicircle** of radius 2 above the x-axis from $x = 0$ to $x = 4$
- A **line segment** from $(4, 0)$ to $(8, -4)$

It is given that $f(0) = 5$.

(a) Find all x-values where f has a local maximum on $(0, 8)$. Justify.

(b) Find $f(4)$.

(c) On what interval(s) is f concave down on $(0, 8)$? Justify.

Show your work — set-up, work, and final answer with units.

Question 35

FRQ · TABLE & RIEMANN

The velocity of a car (in ft/s) is recorded at 5-second intervals:

T (SEC)	0	5	10	15	20
v(t)	30	45	50	55	40

(a) Use a midpoint Riemann sum with **2** equal subintervals to approximate $\int_0^{20} v(t)\,dt$.

(b) Interpret $\int_0^{20} v(t)\,dt$ in context, with units.

(c) Use a trapezoidal sum with the **4** subintervals shown to approximate $\int_0^{20} v(t)\,dt$.

Show your work — set-up, work, and final answer with units.

Submit Your Answers

Click below to copy a clean, formatted version of all your answers. Paste it into Canvas, email, or anywhere else your teacher accepts submissions.

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